

Private Markets Investment Data Lakehouse & Document Intelligence Platform

Portfolio engineering briefing

A synthetic end-to-end financial-services data platform covering ingestion, explicit quality controls, historical correction handling, Data Vault 2.0, Kimball reporting marts, reconciliation, document extraction, CI and a GitHub Pages dashboard.

Metric	Verified local result
Funds / investors / assets	24 / 180 / 240
Valid cashflows / positions	4,547 / 272
Quarantined invalid rows	63
NAV restatements preserved	18
Capital-call PDFs parsed	60
Document-to-ledger review queue	8
Position-to-NAV fund reviews	6
Median synthetic TVPI / DPI / XIRR	1.089x / 0.620x / 6.4%
Automated tests	37/37 passed

Execution boundary

The complete reference build was executed locally with Python, pandas and SQLite. Databricks/PySpark/Delta notebooks are included as deployment assets and were source/syntax checked, but they were not run in a live Databricks workspace. This distinction is deliberately documented so portfolio claims remain defensible.

Architecture and modelling choices

Layer	Purpose
Bronze	Immutable raw source rows plus ingestion metadata.
Silver	Validated, standardised records; invalid rows route to quarantine with an explicit reason.
Data Vault 2.0	Hubs/links/satellites preserve business keys, relationships and NAV restatement history.
Kimball Gold	Fund/investor/asset/date dimensions plus cashflow, commitment, position and NAV facts.
Semantic marts	TVPI/DPI/RVPI, unfunded commitment, exposure, reconciliation and review queues.
Presentation	Self-contained Plotly dashboard suitable for GitHub Pages.

Control design

The generated raw data deliberately contains duplicate cashflows, invalid currencies, orphan keys, negative commitments, invalid NAV values and orphan position assets. The pipeline quarantines those records rather than silently repairing them. NAV restatements are retained historically, while the Gold layer selects the latest valid version for consumption.

Capital-call document control

Sixty synthetic PDF notices are parsed into typed fields and reconciled to the referenced ledger contribution. Eight deliberately mismatched notices are routed to review. The repository includes an optional model-serving seam for a future LLM extraction step, but deterministic validation and reconciliation remain mandatory.

Portfolio relevance and limitations

What this demonstrates: financial-data ingestion, formal warehouse modelling, auditability, data quality, correction handling, business reconciliation, document controls, automated testing, CI and client-facing analytical outputs.

What it does not claim: production deployment, live Databricks execution, real private-markets performance, audited investment calculations or deployed AI extraction. All entities and values are synthetic.

Currency handling: fund-level ratios are calculated in each fund's base currency. Cross-fund monetary totals and exposure marts are normalised to GBP through a validated synthetic FX table, preventing unlike currencies from being added directly.

Next live-platform step: import the numbered notebooks in databricks/notebooks/ into a Databricks workspace, upload the generated inputs to a Unity Catalog volume, execute the Bronze-to-Gold path and only then update the CV wording to claim direct Databricks implementation.